

```
> mkdir my_dir

~/Doc/AUCA/FALL_2026/Information_Security/home
> cd my_dir

~/Doc/AU/FALL_2026/Information_Security/home/my_dir
> touch my_file.txt

~/Doc/AU/FALL_2026/Information_Security/home/my_dir
> chmod 777 my_file.txt

~/Doc/AU/FALL_2026/Information_Security/home/my_dir
> cat my_file.txt

~/Doc/AU/FALL_2026/Information_Security/home/my_dir
> cd ..

~/Doc/AUCA/FALL_2026/Information_Security/home
> mkdir another_dir

~/Doc/AUCA/FALL_2026/Information_Security/home
> cp my_dir/my_file.txt another_dir/

~/Doc/AUCA/FALL_2026/Information_Security/home
> rm my_dir/my_file.txt

~/Doc/AUCA/FALL_2026/Information_Security/home
> cd another_dir

~/Doc/AU/FALL_2026/I/home/another_dir
> ls
my_file.txt
```

**Task: Create two directories and move all files instead of copying from one directory to another**

```
~/Doc/AUCA/FALL_2026/Information_Security/home
> mkdir dir1

~/Doc/AUCA/FALL_2026/Information_Security/home
> mkdir dir2

~/Doc/AUCA/FALL_2026/Information_Security/home
> cd dir1

~/Doc/AUCA/FALL_2026/Information_Security/home/dir1
> touch file1.txt

~/Doc/AUCA/FALL_2026/Information_Security/home/dir1
> touch text.txt

~/Doc/AUCA/FALL_2026/Information_Security/home/dir1
> touch password.txt

~/Doc/AUCA/FALL_2026/Information_Security/home/dir1
> ls
file1.txt      password.txt  text.txt

~/Doc/AUCA/FALL_2026/Information_Security/home/dir1
> mv * ../dir2/

~/Doc/AUCA/FALL_2026/Information_Security/home/dir1
> cd ../dir2

~/Doc/AUCA/FALL_2026/Information_Security/home/dir2
> ls
file1.txt      password.txt  text.txt

~/Doc/AUCA/FALL_2026/Information_Security/home/dir2
```

```
> curl -help
Usage: curl [options...] <url>
  -d, --data <data>          HTTP POST data
  -f, --fail                  Fail fast with no output on
HTTP errors
  -h, --help <category>      Get help for commands
  -i, --include                Include response headers in
output
  -o, --output <file>         Write to file instead of std
out
  -O, --remote-name           Write output to file named a
s remote file
  -s, --silent                Silent mode
  -T, --upload-file <file>    Transfer local FILE to desti
nation
  -u, --user <user:password>  Server user and password
  -A, --user-agent <name>     Send User-Agent <name> to se
rver
  -v, --verbose               Make the operation more talk
ative
  -V, --version               Show version number and quit
```

This is not the full help, this menu is stripped into categories.

Use "--help category" to get an overview of all categories.

For all options use the manual or "--help all".

```
> ~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> |
```

```
> curl https://jsonplaceholder.typicode.com/posts
[
  {
    "userId": 1,
    "id": 1,
    "title": "sunt aut facere repellat provident occaecat
i excepturi optio reprehenderit",
    "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
  },
  {
    "userId": 1,
    "id": 2,
    "title": "qui est esse",
    "body": "est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui aperiam non debitis possimus qui neque nisi nulla"
  },
  {
    "userId": 1,
    "id": 3,
    "title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",
    "body": "et iusto sed quo iure\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit aut"
  },
  {
    "userId": 1,
```

```
> curl -o filename.json https://jsonplaceholder.typicode.com/posts
```

| % Total<br>Time | % Received<br>Time | % Xferd<br>Current | Average<br>Speed | Time<br>Total |      |   |          |    |
|-----------------|--------------------|--------------------|------------------|---------------|------|---|----------|----|
| Spent           | Left               | Speed              | Dload            | Upload        |      |   |          |    |
| 0               | 0                  | 0                  | 0                | 0             | 0    | 0 | --:--:-- | -- |
| 100             | 27520              | 0 27520            | 0                | 0             | 143k | 0 | --:--:-- | -- |
| :--:--          |                    | --:--:--           | 143k             |               |      |   |          |    |

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ls
```

```
filename.json
```

```
> curl -I https://jsonplaceholder.typicode.com/posts
HTTP/2 200
date: Tue, 15 Sep 2026 07:24:43 GMT
content-type: application/json; charset=utf-8
access-control-allow-credentials: true
cache-control: max-age=43200
etag: W/"6b80-Ybsq/K6GwwqrYkAsFxqDXGC7DoM"
expires: -1
nel: {"report_to":"heroku-nel","response_headers":["Via"],
,"max_age":3600,"success_fraction":0.01,"failure_fraction":0.1}
pragma: no-cache
report-to: {"group":"heroku-nel","endpoints":[{"url":"https://nel.heroku.com/reports?s=GZM6aoxxWPVfkQ0GPfQ%2B0iwseZGec0m%2Ff04UTGzUGPs%3D\u0026sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d\u0026ts=1785317901"}], "max_age":3600}
reporting-endpoints: heroku-nel="https://nel.heroku.com/reports?s=GZM6aoxxWPVfkQ0GPfQ%2B0iwseZGec0m%2Ff04UTGzUGPs%3D&sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d&ts=1785317901"
server: cloudflare
vary: Origin, Accept-Encoding
via: 2.0 heroku-router
x-content-type-options: nosniff
x-powered-by: Express
x-ratelimit-limit: 1000
x-ratelimit-remaining: 999
x-ratelimit-reset: 1785317903
age: 10932
cf-cache-status: HIT
cf-ray: a3b5e4b2bf456074-FRU
alt-svc: h3=":443"; ma=86400
```

```
> curl -X POST -H "Content-Type: application/json" \
-d '{"title": "foo", "body": "bar", "userId": 1}' \
https://jsonplaceholder.typicode.com/posts
```

```
{
  "title": "foo",
  "body": "bar",
  "userId": 1,
  "id": 101
}
```

~/Doc/AUCA/FALL\_2026/Information\_Security/home

```
> |
```

**Task: try to discover *tar* command**

```
> tar --help
tar(bsdtar): manipulate archive files
First option must be a mode specifier:
  -c Create  -r Add/Replace  -t List  -u Update  -x Extra
ct
Common Options:
  -b #      Use # 512-byte records per I/O block
  -f <filename> Location of archive
  -v        Verbose
  -w        Interactive
Create: tar -c [options] [<file> | <dir> | @<archive> | -
C <dir> ]
  <file>, <dir> add these items to archive
  -z, -j, -J, --lzma Compress archive with gzip/bzip2/xz
/lzma
  --format {ustar|pax|cpio|shar} Select archive format
  --exclude <pattern> Skip files that match pattern
  -C <dir> Change to <dir> before processing remaining f
iles
  @<archive> Add entries from <archive> to output
List: tar -t [options] [<patterns>]
  <patterns> If specified, list only entries that match
Extract: tar -x [options] [<patterns>]
  <patterns> If specified, extract only entries that mat
ch
  -k        Keep (don't overwrite) existing files
  -m        Don't restore modification times
  -O        Write entries to stdout, don't restore to disk
  -p        Restore permissions (including ACLs, owner, file
flags)
bsdtar 3.5.3 - libarchive 3.7.4 zlib/1.2.12 liblzma/5.4.3
bz2lib/1.0.8
```

```
> mkdir dir2
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ls
```

```
dir2
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> tar -cvf archive.tar dir2
```

```
a dir2
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ls
```

```
archive.tar dir2
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> tar -tvf archive.tar
```

```
drwxr-xr-x  0 arsen  staff          0 Sep 15 13:26 dir2/
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> mkdir extracted
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ls
```

```
archive.tar dir2      extracted
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> tar -xvf archive.tar -C extracted
```

```
x dir2/
```

```
~/Doc/AUCA/FALL_2026/Information_Security/home
```

```
> ls extracted
```

```
dir2
```